Self Study Lab Exercises | Week 1

COSC2779/COSC2972 - Deep Learning

1 Introduction

In this week's lab, we will explore deep learning infrastructure that will be used during the course. The main cloud-based platforms we will be using are:

- **Google Colab**
- AWS env. via RMIT RACE hub - this is very similar to Colab and will only be used for assignments. We will provide a demonstration later.

The following text will provide an introduction to setting up the above platform for training and testing deep neural network models. At the end of each section, there are links to some video tutorials. **It is highly recommended that you go through them to get a good idea of Google Colab**.

2 Google Colab

The first platform we are going to explore is Google Colab. Colab (short for Colaboratory) is a product offered by Google Research that allows machine learning researchers to work on projects in the browser. Similar to Google Docs, it allows you to share projects between many people (for group work), and best of all, it gives *free access to GPUs* for you to quickly train models. On the negative side, Colab sessions are limited to 12 hours (you can restart after 12 hours) and there is no guarantee that you will get a GPU (however, most of the time you will get a GPU).

A nice introduction to Colab is at: https://youtu.be/inN8seMm7UI

In order to start, you need to have a Google account. You can use your **personal Gmail account** (RMIT student email is no longer supported).

Setting up a project

- Login to your preferred Google account and open GDrive.
- Create a folder with a suitable name for the project (e.g., COSC2779lab1) and navigate into that folder.
- Right click → Google Colaboratory; (if you can't see it, click more).
- This will open a notebook instance. You can rename the notebook.
- **If you prefer to use a GPU**: Runtime → Change Runtime Type → Select GPU from the dropdown box under Hardware accelerator.

Now you are ready to use Python on a GPU instance. Using Python on Colab is the same as on a regular Anaconda Jupyter notebook.

Test and install PyTorch

Let's first check if PyTorch is installed, check its version, and verify that it can detect the GPU.

```
import torch
print("PyTorch Version:", torch.__version__)
print("GPU Available:", torch.cuda.is_available())
```

This should output a version number (typically 2.0 or higher) and `True` for the GPU availability if you selected a GPU runtime. If it is not installed, you can install it using the following command. Note that the "!" character allows you to run command-line arguments.

```
!pip install torch torchvision torchaudio
```

Modern Colab instances come with PyTorch pre-installed, so it is rarely necessary to use the pip command to install it. The above line is just to indicate that you can install Python packages in the notebook using the command line.

Uploading Files

Any DL project will involve a significant amount of data, and the first step is to get them into the compute instance. There are several methods to get data into a Colab instance, and here we will discuss the two most popular methods.

Uploading from GDrive

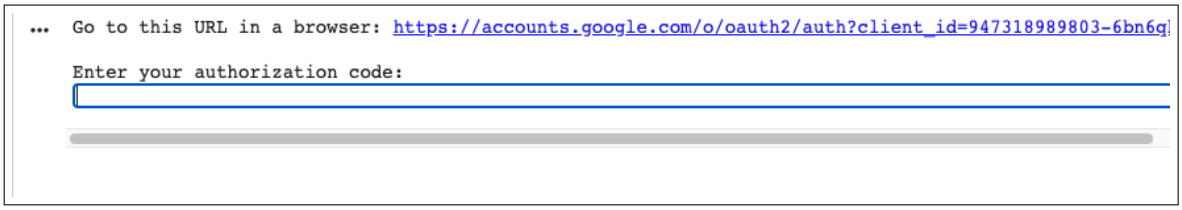
Assume our data is available in GDrive as a zip file. For this section, you can download the "*BostonHousingPrice.zip*" file from Canvas and upload it to the folder created earlier (e.g., COSC2779lab1). This zip file contains a *.csv* file with Boston house price data, including house prices and some factors associated with the prices to try to predict house prices.

First, you need to mount your GDrive onto the instance. The easiest way to do this is by clicking the **Folder icon** on the left sidebar and clicking the **Mount Drive** button.

Alternatively, this can be done by running the following code block in the notebook:

```
from google.colab import drive
drive.mount('/content/drive')
```

This may ask for an authentication code or prompt a popup window for you to log in to your Google account:



Follow the prompts to obtain the authentication code or allow access. Now your GDrive is mounted. You can check your files by:

```
!ls /content/drive/'My Drive'/COSC2779/COSC2779lab1/
```

Working with data directly on GDrive might be too slow. Especially when working with large datasets that will be read one batch at a time (typical for deep learning), reading data directly from Drive during training creates a network bottleneck that will drastically slow down your GPU. Therefore, we will copy the data to the local disk of the notebook instance using the following code.

```
!cp /content/drive/'My Drive'/COSC2779/COSC2779lab1/BostonHousingPrice.zip .
!unzip -q -o BostonHousingPrice.zip
!rm BostonHousingPrice.zip
!ls
```

Now we can read the data using pandas and use it for our ML model development:

```
import pandas as pd
import numpy as np

data = pd.read_csv('./BostonHousingPrice/housing.data.csv',delimiter='\s+')
data.head(5)
```

Direct Upload from your computer

The easiest way to get your files into the notebook instance is to open the left sidebar (Folder icon) and **drag-and-drop** files directly from your computer into the Colab file tree.

Alternatively, you can upload them using Python code. This method is less recommended for large files as it can be time-consuming, and you will need to re-upload data every time you reinitialize the notebook instance (Factory reset).

First, do a Runtime → Factory reset runtime. This will remove all the data you uploaded in the previous section. Then you can upload the code, extract the zip file, and read the dataframe using the following code.

```
import pandas as pd
import numpy as np
# The following lines will prompt an upload widget in the notebook
from google.colab import files
files.upload()
# The lines below are the same as in the previous section
!unzip -q -o BostonHousingPrice.zip
!rm BostonHousingPrice.zip

data = pd.read_csv('./BostonHousingPrice/housing.data.csv',delimiter='\s+')
data.head(5)
```

! You are now setup to use Colab. For more information, please follow the tutorials at:

- Video tutorial for using PyTorch on Colab
- Section: Working with Notebooks in Colab & VSCode Extension